

Surname and Initials:	Session Day:	Group no
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UNIVERSITEIT VAN PRETORIA
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DEPARTMENT OF PHYSICS

FSK 116

Practical Report

COEFFICIENT OF STATIC FRICTION BETWEEN SHEETS OF PAPER

Part 1: Background Questions

Question 1: Does the static friction between two surfaces depend on the size of the surfaces? Give a reason for your answer.

Question 2: Consider a stationary object of mass m on a rough surface inclined at an angle θ . Draw a labelled free-body diagram of all the forces acting on the body and include an appropriate set of axes.

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Question 3. Using Newton’s laws, find different expressions for the magnitude of the normal force using both the x -direction and the y -direction of the chosen set of axes.

Question 4. From the two expressions for the normal force magnitude, find the relationship between the static coefficient of friction, μ_s and the angle of inclination, θ .

Part 2: Experiment

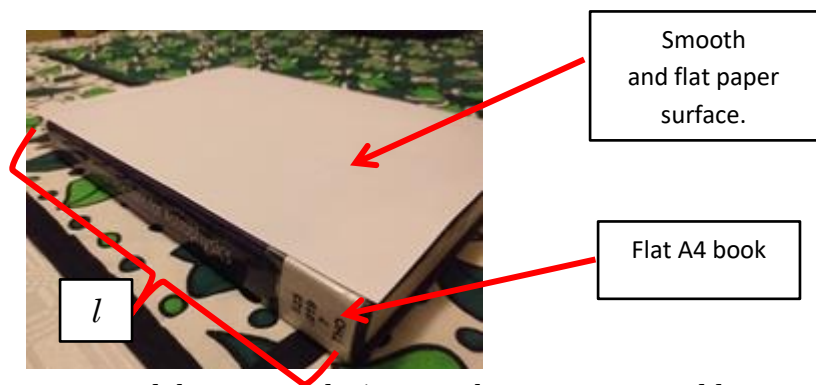
AIM: Determine the static coefficient of friction between two sheets of standard paper.

Experimental apparatus:

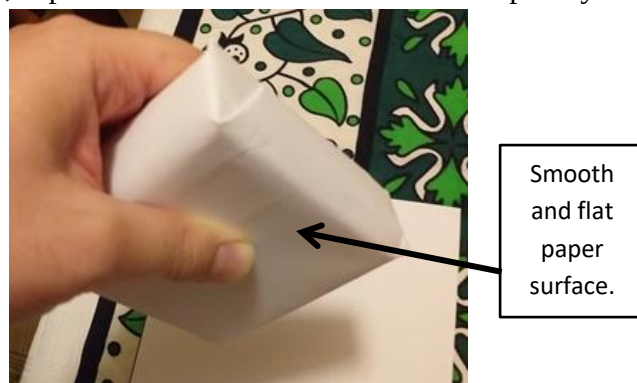
1. Two standard sheets of writing paper. Pages must not be glossy, must be clean, free of dust and must be the same. Pages may have writing on. Pages from an exam pad or standard printing paper are ideal. Newspaper is acceptable, if nothing else can be found.
2. A ruler for measuring length.
3. A large, flat, hard-cover book (about A4) or a clipboard or piece of hardboard of a similar size.
4. A smaller, rectangular or square object about the size of a phone. A flat phone can be used.

Experimental procedure:

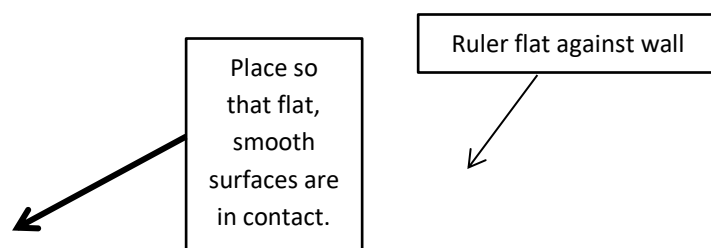
1. Measure the length of the large, flat A4 book (or clipboard, etc).
2. Place one sheet of paper on the large, flat A4 flat book (or clipboard, etc). If possible, stick it in place. Make sure surface is smooth and completely flat.



3. Wrap the second piece of paper around the rectangular/square object, as you would wrap a present. Stick paper in place, if possible. Make sure one side is completely smooth and flat.



4. Choose a well-lit room in your home. At a point where the wall meets the floor, place the ruler vertically against the wall. Place the flat A4 book (or clipboard, etc.) with the paper, flat on the floor, next to the ruler.





5. Place the smooth side of the wrapped rectangular/square object roughly in the middle of the paper on the flat A4 book (or clipboard, etc.).
6. Show your experimental setup (take a picture of your experimental setup)

INSERT PICTURE OF EXPERIMENTAL SETUP HERE

7. Carefully and slowly lift the one side of the flat A4 book (or clipboard, etc.) so that it forms an incline. The other end must still be in contact with the floor. Just at the moment the wrapped rectangular/square object starts moving, make a note of the height above the floor of the lifted end of the book.
8. Repeat 20 times, recording the height each time. Tabulate your results. Remember to write units where applicable.

Title:

Calculations:

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- A right-angled triangle is shown. The vertical side is labeled h . The hypotenuse is labeled l . The angle at the bottom right vertex is labeled θ .

2. Using the tabulated results and the measured length of the large, flat A4 book (or clipboard, etc.), determine the 20 values for the coefficient of static friction, μ_s using the result of the equation found in question 4 from Part 1. Show an example of a full calculation of μ_s below and then tabulate the rest of the results in table 2.

Example of calculations:

Table 2

Title:

[illegible]

3. Determine the average static coefficient of friction. Compare this with the values you may be able to find from literature. Remember to cite the literature properly. Comment on your result: do you think it is reliable?

4. Suggest two or more improvements to the experiment.

5. Comment on why studying friction between materials may be useful

END